Technological Project,

Markowitz's Model of Risk

Andal Iman - 55892200
Guyot Gaspard - 57902200
Houart Simon - 5778220
Houriez Basile - 39502100
Lebbe Cyril - 89332100
Philippart Emilien - 28072200
Poncin Nicolas - 76152100

Groupe_OR2

1. Introduction and project objectives

This Operational Research project aims to model an efficient financial portfolio based on a sample of companies from the BEL20 index, using the Markowitz model. The work involves calculating the expected returns and the variance-covariance matrix for 20 assets. The objective is to construct the efficient frontier under three scenarios: firstly, with no restrictions, secondly, with a portfolio containing only long positions, and lastly, with a portfolio containing only 5 assets. The goal is therefore to minimize portfolio variance for a given level of expected return across all three scenarios.

2. Calculation of returns, variance, and Covariance Matrix

Using historical weekly prices of BEL20 companies, returns were calculated as follows:

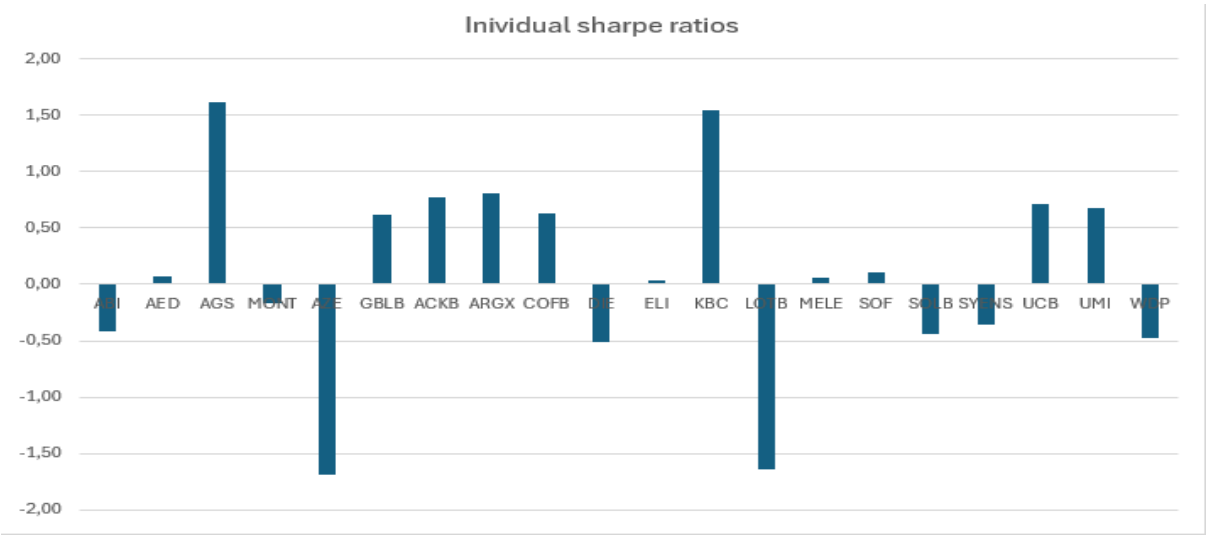
$$Ri = \frac{Pt - Pt-1}{Pt-1}$$

We then calculated the standard deviation of these returns. Subsequently, the data were annualized by multiplying returns by 52 and standard deviations by $\sqrt{52}$.

A correlation heatmap was generated, revealing varying relationships between assets, ranging from -0.49 (ARGX and ABI) to +0.7 (MONT and WDP).

We also calculated the individual Sharpe ratio for each asset in order to assess their risk-adjusted performance. This ratio measures the excess return per unit of risk, allowing for comparison across assets with different risk profiles. It was calculated as follows:

$$Sharpe = \frac{R_i - R_f}{\sigma_i}$$



Finally, we constructed the covariance matrix (see Python code):

$$\Sigma = E\left[(R-\mu)(R-\mu)'\right]$$

This matrix captures the co-movement of assets: diagonal elements represent individual asset variances, while off-diagonal elements reflect correlations between assets. This structure is crucial as low covariance between two assets enhances diversification benefits, whereby positive contributions from another offset negative effects from one asset on the portfolio.

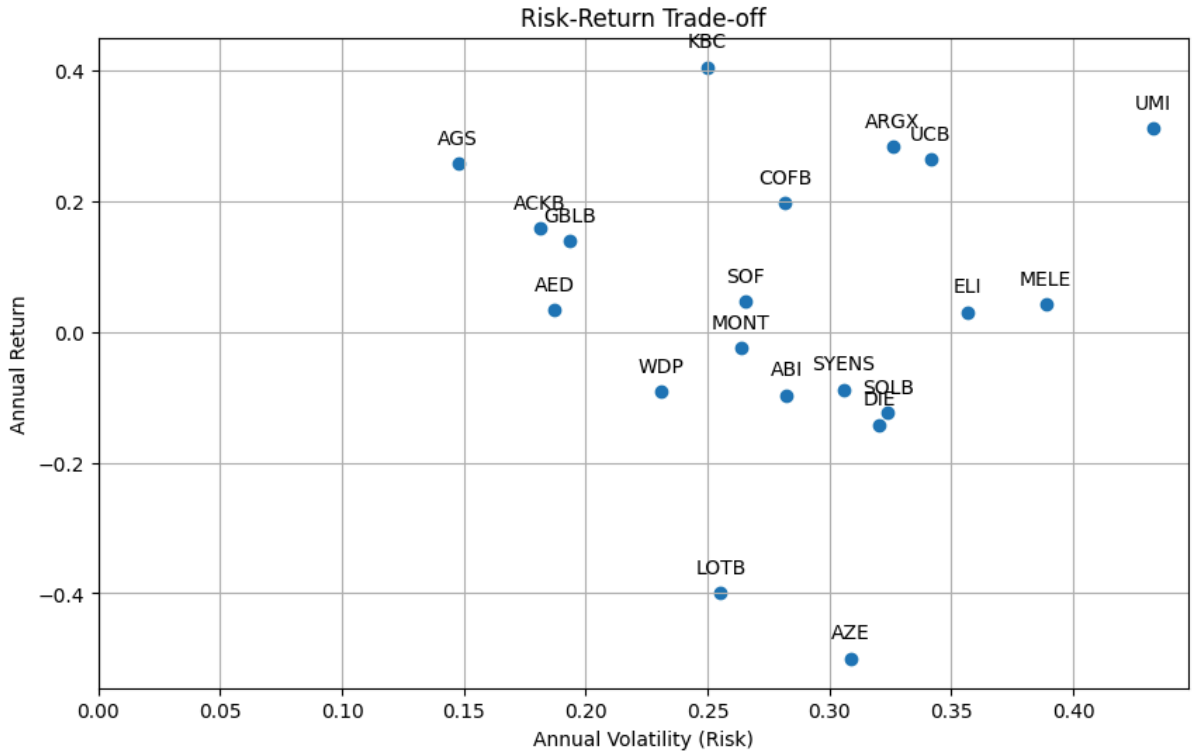
3. Efficient Frontier Modeling

The Markowitz model aims to minimize portfolio variance for a target return level. The optimization problem can be formulated as follows:

$$\min_{w} \quad w'Vws.c. \quad w'\mathbf{1} = 1, w'R = R^*, w_i \geq 0 \quad (long-only)$$

where w represents the vector of portfolio weights, V is the covariance matrix, μ denotes the vector of expected returns, and R* is the target return.

We first plotted a risk-return trade-off map to visualize the positioning of individual stocks.



The upper-left corner represents the most desirable region, as it minimizes risk while maximizing returns. Building upon this plot, we added the efficient frontier for each scenario,

along with two key portfolios: the minimum variance portfolio (MVP), representing the lowest possible risk and positioned at the leftmost point of the efficient frontier, and the tangency portfolio, which represents the optimal risk-return combination by maximizing the Sharpe ratio—the point where a line from the risk-free rate is tangent to the efficient frontier.

Three distinct optimizations were performed using SLSQP (Sequential Least Squares Programming) in Python:

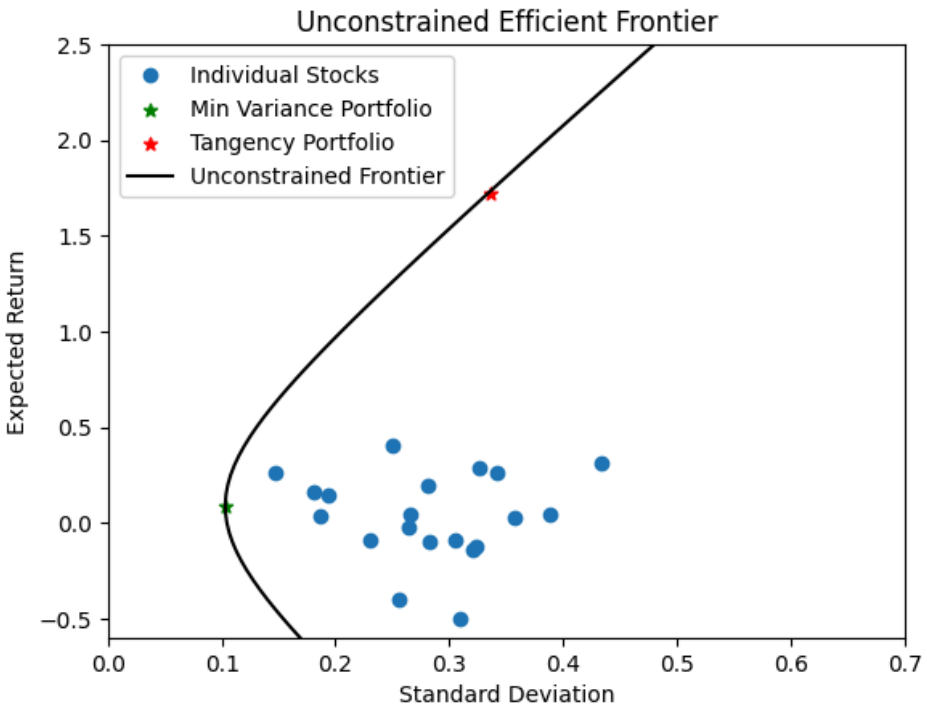
a) Scenario 1: Unconstrained Portfolio

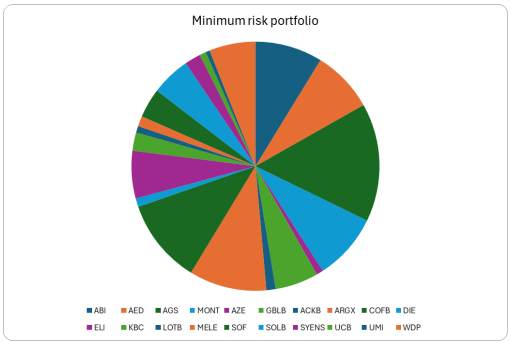
For the unconstrained scenario, we constructed the efficient frontier by solving the portfolio optimization problem (across a range of target returns from -1 to 3, with increments of 0.05).

For each target return level, the algorithm minimizes portfolio variance subject to two equality constraints:

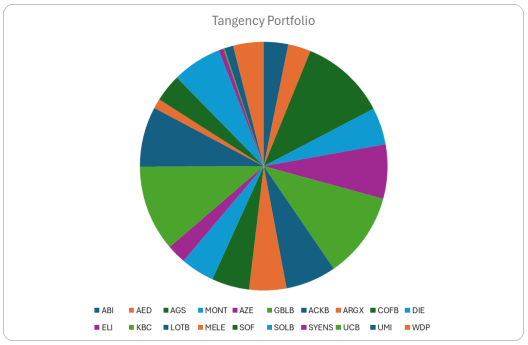
- the sum of portfolio weights equals one (full investment)
- the portfolio's expected return is positive. (otherwise it minimizes the Sharp ratio negatively

The resulting efficient frontier was plotted alongside individual stock positions in risk-return space. Each point on the frontier represents an optimal portfolio for its corresponding return level, demonstrating the minimum achievable volatility (standard deviation) for each target return. Two stars on the frontier highlight key portfolios: the minimum variance portfolio (leftmost point) and the tangency portfolio (maximizing the Sharpe ratio).





Minimum variance portfolio: expected return of 8.91%, volatility of 10.32%



Tangency portfolio: expected return of 171.69%, volatility of 33.60

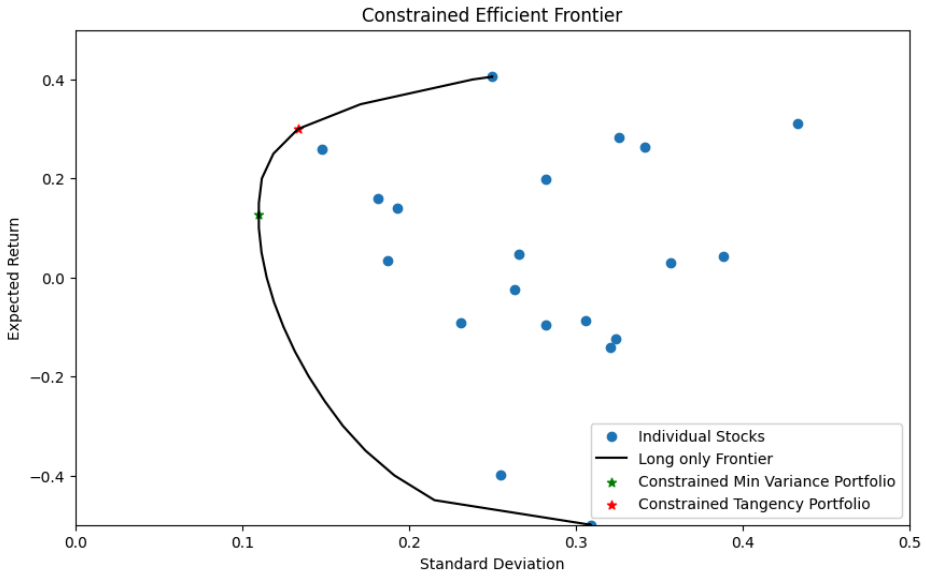
However, the tangency portfolio obtained requires excessive leverage, making the solution impractical. This occurs because our linear constraints allow unrestricted weight magnitudes. To address this limitation, we should implement a quadratic constraint, such as: $-1 < w_i < 1$

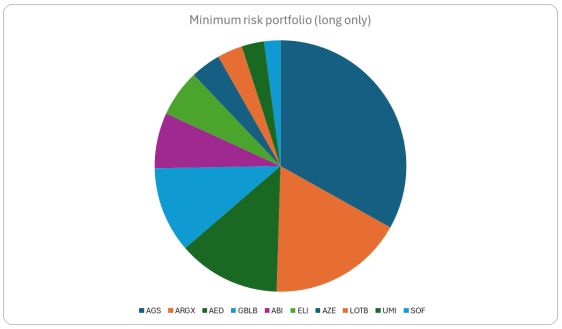
b) Scenario 2: Long-Only Portfolio

In this scenario, we maintained the same optimization settings while adding the non-negativity constraint, $w_i \geq 0$ which excludes all negative weights, ensuring that each asset is held exclusively in long positions. The optimization is subject to three constraints:

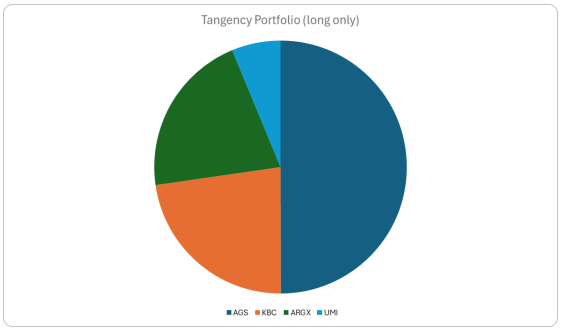
- The sum of portfolio weights equals one (full investment)
- The portfolio's expected return is positive (otherwise the Sharpe ratio would be minimized negatively)
- No short selling: each individual weight must be non-negative

Consequently, the efficient frontier becomes shorter and more convex, positioned below the unconstrained version. Diversification is more limited, as high-return but highly volatile assets can no longer be offset by short positions.





Minimum variance portfolio: expected return of 12.74%, volatility of 10.98%



Tangency portfolio: expected return of 30.04%, volatility of 13.37%

c) Scenario 3: Restricted Portfolio (bonus)

For this bonus question, we approached each problem separately rather than developing a unified model. The newly introduced constraint added quadratic complexity that disrupted our optimization framework, preventing us from finding a solution limited to exactly 5 assets.

However, for the tangency portfolio, we applied a similar methodology and successfully obtained results. Since our original tangency portfolio (long-only, without the 5-asset restriction) naturally concentrated in only 4 assets, it's unsurprising that adding the maximum 5-asset constraint yielded an identical portfolio composition.